

POST-PRUNING FINE-TUNING GAINS DEPEND ON THE TEMPORAL EVALUATION OPERATING POINT IN AUDIOLDM

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ABSTRACT

Structured pruning of a text-to-audio diffusion model is followed by recovery fine-tuning, whose benefit is usually summarised by a single aggregate score at one inference setting. Studying the publicly released pruned and post-fine-tuning checkpoints of AudioLDM-M-Full, we show that this benefit depends strongly on the temporal evaluation operating point. With a paired, common-noise design we define a temporal interaction J : the fine-tuning advantage at the native 10.24s scale minus that at a shorter 3.84s scale. J is borderline at a milder pruning severity and is prospectively confirmed at a stronger severity on a disjoint set ($J = +0.159$, 95% CI $[+0.131, +0.187]$), robust to the pruning-seam convention. The advantage stays positive at the short point (not a 10s-only effect), and an independent frame-level model shows the native gain is temporally broad. Post-pruning recovery should therefore be reported across inference operating points, not collapsed to one score; lacking a matched dense fine-tuned control or human listening, the claim is about evaluation, not mechanism.

Index Terms— text-to-audio generation, diffusion models, structured pruning, fine-tuning, evaluation

1. INTRODUCTION

Structured pruning followed by recovery fine-tuning is a standard route to cheaper generative diffusion models [3, 4], including text-to-audio (TTA) generation with AudioLDM [1, 2]. The effect of the recovery stage is normally reported as the restoration of one aggregate benchmark number—a Fréchet Audio Distance or a KL divergence on an in-domain test set at a fixed sampler setting.

A generative model, however, is queried at an inference operating point—clip duration, sampler steps, guidance—and there is no guarantee that a single aggregate number characterises the fine-tuned checkpoint at every operating point a user might choose. We ask a narrow, empirical question: does the advantage of a post-pruning fine-tuned checkpoint over the pruned checkpoint depend on the temporal evaluation operating point? We study the publicly released pruned and post-fine-tuning checkpoints of AudioLDM-M-Full, treating them as the object of study rather than proposing a new pruning or fine-tuning method. The contribution is therefore an evaluation:

- a paired, common-noise evaluation framework that isolates the post-fine-tuning—pruned contrast at two temporal operating points and a temporal interaction J

that quantifies how that contrast changes with duration;

- a prospectively confirmed interaction at a stronger pruning severity on a disjoint set, robust to a pre-specified pruning-seam convention;
- a multi-metric and frame-level characterisation showing the native-scale advantage is broad but heterogeneous, with explicit boundary conditions on what can and cannot be concluded.

The originally hypothesised out-of-domain specialisation effect does not survive testing; we report it in Section 4 as the step that motivated the operating-point question, and keep every other negative.

2. RELATED WORK

Diffusion pruning and recovery. Structural pruning of diffusion U-Nets with fine-tuning recovery has been studied for image models [3, 4] and, for TTA, by Singh et al. [2], who L1-prune an AudioLDM-family U-Net and recover on AudioCaps, reporting in-domain FAD/KL at a fixed sampler. These works evaluate recovery at one operating point; we vary it and measure the change in the fine-tuning advantage.

Compression and inference regime. We are not aware of prior work establishing, for structural pruning of a TTA model, an evaluation-time interaction between the recovery advantage and the temporal operating point; compression error is known to interact with the sampler step count.

TTA evaluation. CLAP similarity [5] is a standard automatic proxy for text–audio alignment; distributional metrics such as FAD [7] and event classifiers such as PANNs [6] are common but sample-size and encoder sensitive. These limits motivate a paired, multi-metric treatment.

3. METHOD

3.1. Systems and pruning severities

The dense model is AudioLDM-M-Full [1], a latent-diffusion U-Net with stage multipliers `channel_mult = (1, 2, 3, 5)` (415.96 M U-Net parameters), itself AudioCaps-specialised by the original authors. We use the publicly released L1 structurally pruned checkpoints at two severities that prune the deeper U-Net stages more aggressively (Table 1): a milder severity 1 (1, 2, 3, 1) and a stronger severity 2 (1, 2, 1, 1), reducing the U-Net by 65.0% and 82.9% respectively. We verified that our pruning operator reproduces the released

Table 1. U-Net size of AudioLDM-M-Full at each pruning severity. Parameter counts are exact, summed over the diffusion-model tensors of the materialised checkpoints.

System	channel_mult	U-Net	Reduction
Dense	(1, 2, 3, 5)	415.96 M	—
Severity 1	(1, 2, 3, 1)	145.67 M	−65.0%
Severity 2	(1, 2, 1, 1)	71.08 M	−82.9%

pruned weights bit-exactly (per-tensor oracle checks) and we train no system.

3.2. Post-pruning fine-tuning checkpoints

The post-fine-tuning checkpoint at each severity is the released checkpoint obtained by fine-tuning the pruned U-Net on AudioCaps for 10^6 steps [2]; we load it, materialise its exponential-moving-average weights, and verify its checksum. “Recovery” names the AudioCaps fine-tuning stage, not an achieved restoration.

3.3. Batteries and operating points

The in-domain battery is AudioCaps [8] test captions; the out-of-domain battery is a held-out set of hip-hop-style captions derived from a MusicCaps-style source [9], with near-zero exposure to that subdomain in the recovery corpus. We generate at a short point (3.84s, latent length 96) and the fine-tuning-native native point (10.24s, latent length 256); the music battery is generated only at the short point. All generation uses DDIM [10] with 50 steps, guidance 2.5, $\eta = 0$, fp32, single generation, EMA weights. The severity-2 sets were constructed after the severity-1 results and are disjoint from them: 192 AudioCaps prompts (excluding severity-1 and training prompts) and 64 new music prompts. Severity 1 uses 96 AudioCaps prompts for the controlled test, a matched 80-prompt subset for the temporal comparison, and 64 music prompts.

3.4. Paired design, scoring, statistics

For each prompt and operating point the compared systems receive identical generation noise x_T (common random numbers), so a per-prompt contrast is not confounded by sampling noise. The primary scorer is CLAP (laion/clap-ltsat-fused, rev. 365dea6e) [5], applied with a fixed batch/order/seed convention that is part of the endpoint. The unit is the prompt (music replicates averaged first); intervals are 95% prompt-level percentile bootstrap ($B = 10^4$).

3.5. Estimands, gates, evidence classes

Let $R(\text{context})$ be the mean per-prompt CLAP cosine of the post-fine-tuning checkpoint minus the pruned checkpoint; $R_{\text{nat}}, R_{\text{short}}, R_{\text{music}}$ denote the AudioCaps-native, AudioCaps-short and held-out-music contrasts. The central quantity is whether the advantage changes with temporal scale, the temporal interaction

$$J = R_{\text{nat}} - R_{\text{short}}, \quad (1)$$

estimated as a paired per-prompt contrast; a pre-specified gate declares it resolved when $\log_5(J) > 0$. We also report a context contrast $K = R_{\text{nat}} - R_{\text{music}}$; because it changes both the prompt domain and the duration, we read K as a bundled context effect, not a pure-domain effect. Sign gates use a smallest effect size of interest SESOI = 0.025; a short-point equivalence test asks whether the 90% CI of R_{short} lies within $[-0.025, +0.025]$. Epistemically: the severity-2 replication and its seam sensitivity are prospectively frozen primary (protocol/gates fixed before scoring), the severity-1 comparison a prospectively specified follow-up, the second scorer and KL/event metrics secondary corroboration, and the frame-level analysis a post-result diagnostic. Cross-severity magnitude comparisons are exploratory and confounded, and never gate.

4. RESULTS

4.1. Severity 2: a prospective replication

We tested the operating-point pattern prospectively at the stronger severity 2 on the disjoint sets, with estimands and gates frozen before any severity-2 output was scored. The interaction is clearly resolved, $J = +0.159$ [$+0.131, +0.187$], driven by a large native-duration advantage, $R_{\text{nat}} = +0.244$ (Fig. 1b, Table 2). At the native point the pruned checkpoint scores only 0.055 and the post-fine-tuning checkpoint 0.299, so recovery matters most where pruning hurts most. The context contrast is also resolved, $K = +0.235$. Both J and K are robust to the pruning-seam convention: an alternative baseline reproducing the released checkpoint’s decoder-seam convention (differing in three tensors) gives $J = +0.161$ and $K = +0.238$ with the same gate outcomes. The seam alternative is a pre-specified sensitivity analysis and never rescues a failed primary gate.

4.2. The advantage is present, not absent, at the short point

The interaction does not mean “fine-tuning helps only at 10s.” At severity 2 the short-point advantage is itself resolved and positive, $R_{\text{short}} = +0.085$, and the equivalence test fails (90% CI [$+0.069, +0.102$], outside ± 0.025): the advantage is smaller at the short point, not gone. The specific severity-1 music-negative pattern also does not replicate: at severity 2 the music contrast is approximately null, $R_{\text{music}} = +0.009$ (Table 2, Fig. 2a). Thus the temporal interaction replicates prospectively, the bundled context contrast K is also positive, while the exact severity-1 music-negative sign does not replicate.

4.3. Severity 1: the path, and a borderline interaction

The severity-2 test did not come first. A severity-1 music observation had motivated a strict hypothesis—that fine-tuning trades in-domain for out-of-domain alignment (a recovery reversal). Tested prospectively at the matched short point, with hypothesis, estimand and gate fixed before scoring, it fails: the in-domain contrast is essentially zero, $R_{\text{AC}} = -0.002$ ($n = 96$; Table 2), so there is no in-domain alignment to sacrifice, while held-out music is negative, $R_{\text{music}} = -0.094$. This reframed the work from a mechanism claim to an operating-point evaluation. Across

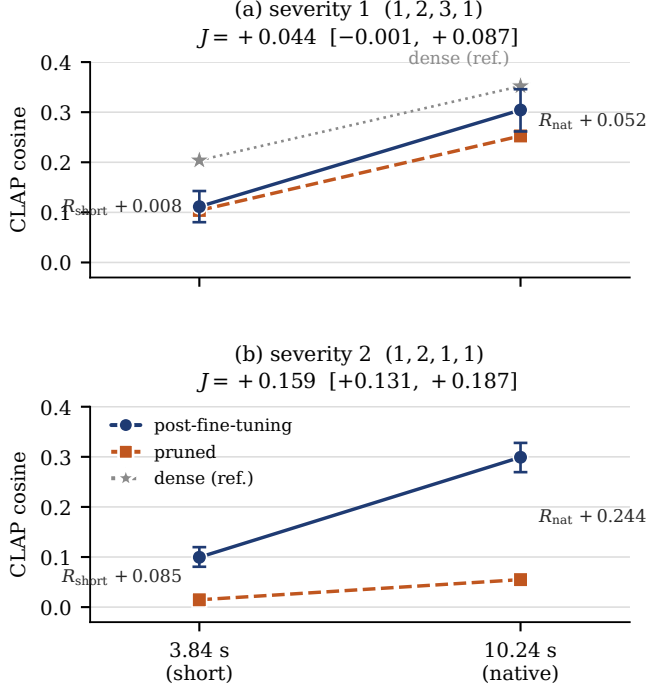


Fig. 1. The fine-tuning advantage grows with the temporal operating point. Mean CLAP cosine of the pruned (dashed) and post-fine-tuning (solid) checkpoints at the short (3.84s) and native (10.24s) points, for (a) severity 1 and (b) severity 2. Whiskers are the 95% CI of the paired contrast R , drawn about the post-fine-tuning mean; the interaction $J = R_{\text{nat}} - R_{\text{short}}$ (with CI) is annotated per panel. The advantage is present at both points but larger at the native scale. The dense model (grey, panel a; a descriptive reference, unpaired) also scores far higher at the native scale, i.e. the short point is off-native for the whole family.

duration on the matched 80-prompt subset the short advantage is near zero ($R_{\text{short}} = +0.008$) and the native advantage resolved ($R_{\text{nat}} = +0.052$), a positive but borderline interaction $J = +0.044 [-0.001, +0.087]$ (CI narrowly includes zero; directional, Fig. 1a)—which motivated the prospective severity-2 confirmation.

4.4. Corroboration

We use secondary metrics conservatively; CLAP is the primary endpoint. At the severity-2 native point a second CLAP-family scorer, Human-CLAP [11] (a CLAP model fine-tuned on human perceptual scores—an automatic model, not a listening test), agrees on a large advantage ($R_{\text{nat}} = +0.375 [+0.340, +0.408]$, $J = +0.185$); on held-out music it shows a small negative contrast ($R_{\text{music}} = -0.037$) that the primary CLAP does not, with no gate role, so the music sign remains unestablished. Two pre-specified event-level metrics implemented after the primary result also favour the post-fine-tuning checkpoint with paired CIs excluding zero: a lower KL to real references (2.23 vs. 4.45; $\Delta = +2.22 [+1.93, +2.53]$) and a higher count of PANN top-10 captured ground-truth labels (1.46 vs. 0.60;

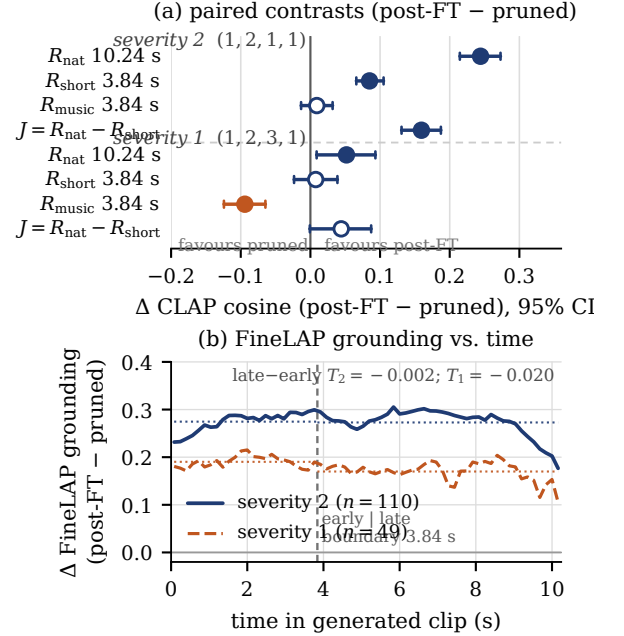


Fig. 2. (a) Paired contrasts R (post-fine-tuning – pruned) with 95% CI; filled markers exclude 0, open markers include 0. At severity 2 the native and short advantages and the interaction J are resolved while R_{music} is null; at severity 1 R_{music} is negative (K bundles domain and duration, see Table 2). (b) FineLAP frame-level grounding gain vs. time in the 10.24s clip (dotted segments: window means $D_{\text{early}}, D_{\text{late}}$; vertical line: 3.84s early/late boundary). The native-scale gain is broad across time, not back-loaded ($T = D_{\text{late}} - D_{\text{early}} \approx 0$); descriptive reconstruction of the frozen diagnostic, no smoothing.

$\Delta = +0.86 [+0.70, +1.02]$). Distribution-level FAD (6.92 vs. 27.4) and FD (48.7 vs. 107.2) point the same way but are single scalars per system without paired CIs, so we report them as descriptive.

4.5. Frame-level diagnostic

As a prospectively frozen post-result diagnostic we score the native 10.24s clips with FineLAP [12], a non-CLAP frame-level language–audio grounding model built on an EAT audio encoder, which reports where in the clip a requested event is grounded. The post-fine-tuning checkpoint shows a large increase in frame-level grounding across the whole clip (semantic mass $+0.27 [+0.22, +0.33]$, occupancy $+0.27$, quarter-coverage $+0.38$, peak $+0.41$; CIs exclude 0; both severities; seam-robust). Crucially the gain is not concentrated late: the late-minus-early difference is essentially zero, $T = -0.002 [-0.024, +0.020]$ at severity 2 (gate fails; -0.020 at severity 1), so the improvement is temporally broad (Fig. 2b). This is frame-level grounding evidence, not calibrated event probability, local causal activation, or perceptual quality.

Table 2. Primary CLAP evaluation: absolute mean cosine of each checkpoint and the paired contrast R = post-fine-tuning – pruned, 95% CI. †: CI excludes 0. ‡: frozen gate $\log_{95} > 0$ passes. Positive favours the post-fine-tuning checkpoint.

Sev.	Setting	Pruned	Post-FT	R [95% CI]
1	3.84 s	0.104	0.111	+0.008 [-0.023, +0.039]
	10.24 s	0.253	0.304	+0.052 [†] [+0.009, +0.093]
	$J = R_{\text{nat}} - R_{\text{short}}$			+0.044 [-0.001, +0.087]
2	3.84 s	0.015	0.100	+0.085 [†] [+0.066, +0.105]
	10.24 s	0.055	0.299	+0.244 [†] [+0.215, +0.273]
	music	0.005	0.014	+0.009 [-0.013, +0.032]
	$J = R_{\text{nat}} - R_{\text{short}}$			+0.159 [‡] [+0.131, +0.187]
	$K = R_{\text{nat}} - R_{\text{music}}$			+0.235 [‡] [+0.197, +0.272]

n : sev. 1 AudioCaps 80 (matched), music 64; sev. 2 AudioCaps 192, music 64. Prospective sev. 1 reversal test (matched 3.84 s, $n=96$): $R_{\text{AC}} = -0.002$ [-0.027, +0.021], gate fails; sev. 1 music $R_{\text{music}} = -0.094$ [-0.124, -0.065]. K bundles domain and duration.

5. DISCUSSION

In our automatic evaluation the post-pruning fine-tuning advantage is not a single number: it depends on the temporal operating point and the evaluation context, and it is heterogeneous across prompts (64% favour post-fine-tuning at the native point vs. 44% at the short point; some favour the pruned one). Recovery should therefore not be treated as a checkpoint-intrinsic scalar. Concretely, we recommend that post-pruning recovery studies for generative models report: (1) the native or training-matched inference operating point; (2) at least one scientifically motivated off-native operating point; (3) absolute pruned and post-fine-tuning performance at each point; (4) paired post-fine-tuning–pruned contrasts using common random numbers where generation is stochastic; (5) uncertainty at the prompt/sample unit, not over generated replicates treated as independent; (6) an explicit interaction contrast across operating points; and (7) a clear separation between automatic semantic metrics and human perceptual evidence. We state this as an implication motivated by this case study, not a universal law: it covers one model family, two pruning severities and one temporal axis.

6. LIMITATIONS

Mechanistic attribution is blocked: the ideal control—a dense, unpruned model given a matched AudioCaps fine-tune—is unavailable (confirmed deleted), and the public dense text-fine-tuned release is not equivalent, so we cannot attribute the dependence to pruning specifically and make no restoration-to-dense claim (the dense reference in Fig. 1a neither resolves a residual gap nor establishes equivalence). Only two temporal points were tested, so no continuous duration trend is established; the short point is off-native and may affect other checkpoints differently. Primary inference relies on CLAP-family semantic metrics (the second scorer is CLAP-derived) and there is no human evaluation (a designed listening study was cancelled pre-launch, zero participants). The operating point differs from the published recipe (DDIM 50 vs. 200), so external validity to it is untested (both

systems are scored identically). The cross-severity magnitude comparison is exploratory and confounded, and the frame-level diagnostic is a post-result grounding measure, native-duration only.

7. CONCLUSION

For the released pruned and post-fine-tuning checkpoints of AudioLDM-M-Full, the fine-tuning advantage depends on the temporal evaluation operating point: an interaction borderline at a milder severity is prospectively confirmed at a stronger severity on a disjoint set, robust to the seam convention, present at the short point, and corroborated by an independent frame-level measure that shows a temporally broad gain. Post-pruning recovery is better described as multi-operating-point behaviour than as a single restored score. Companion audio examples are online.¹

8. REFERENCES

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¹<https://gbibbo.github.io/audioldm-modality-swap-pruning/>